

Block design - Wikipedia

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$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

This article is about block designs with fixed block size (uniform). For block designs with variable block sizes, see [Combinatorial design](#). For [experimental designs](#) in [statistics](#), see [randomized block design](#).

"BIBD" redirects here. For the airport in Iceland, see [Bíldudalur Airport](#).

In [combinatorial mathematics](#), a **block design** is an [incidence structure](#) consisting of a set together with a [family of subsets](#) known as *blocks*, chosen such that number of occurrences of each element satisfies certain conditions making the collection of blocks exhibit [symmetry](#) (balance). Block designs have applications in many areas, including [experimental design](#), [finite geometry](#), [physical chemistry](#), [software testing](#), [cryptography](#), and [algebraic geometry](#).

Without further specifications the term *block design* usually refers to a **balanced incomplete block design (BIBD)**, specifically (and also synonymously) a **2-design**, which has been the most intensely studied type historically due to its application in the [design of experiments](#). Its generalization is known as a **t-design**.

Overview

A design is said to be *balanced* (up to t) if all t -subsets of the original set occur in equally many (i.e., λ) blocks. When t is unspecified, it can usually be assumed to be 2, which means that each *pair* of elements is found in the same number of blocks and the design is *pairwise balanced*. For $t = 1$, each element occurs in the same number of blocks (the *replication number*, denoted r) and the design is said to be *regular*. A block design in which all the blocks have the same size (usually denoted k) is called *uniform* or *proper*. The designs discussed in this article are all uniform. Block designs that are not necessarily uniform have also been studied; for $t = 2$ they are known in the literature under the general name [pairwise balanced designs](#) (PBDs). Any uniform design balanced up to t is also balanced in all lower values of t (though with different λ -values), so for example a pairwise balanced ($t = 2$) design is also regular ($t = 1$). When the balancing requirement fails, a design may still be *partially balanced* if the t -subsets can be divided into n classes, each with its own (different) λ -value. For $t = 2$ these are known as **PBIBD(n) designs**, whose classes form an [association scheme](#).

Designs are usually said (or assumed) to be *incomplete*, meaning that the collection of blocks is not all possible k -subsets, thus ruling out a trivial design.

Block designs may or may not have repeated blocks. Designs without repeated blocks are called *simple*, in which case the "family" of blocks is a [set](#) rather than a [multiset](#).

In [statistics](#), the concept of a block design may be extended to *non-binary* block designs, in which blocks may contain multiple copies of an element (see [blocking_\(statistics\)](#)). There, a design in which each element occurs the same total number of times is called *equireplicate*, which implies a *regular* design only when the design is also binary. The incidence matrix of a non-binary design lists the number of times each element is repeated in each block.

Regular uniform designs (configurations)

The simplest type of "balanced" design ($t = 1$) is known as a **tactical configuration** or **1-design**. The corresponding [incidence structure](#) in [geometry](#) is known simply as a **configuration**, see [Configuration \(geometry\)](#). Such a design is uniform and regular: each block contains k elements and each element is contained in r blocks. The number of set elements v and the number of blocks b are related by $bk = vr$, which is the total number of element occurrences.

Every [binary matrix](#) with constant row and column sums is the [incidence matrix](#) of a regular uniform block design. Also, each configuration has a corresponding [biregular bipartite graph](#) known as its incidence or [Levi graph](#).

Pairwise balanced uniform designs (2-designs or BIBDs)

Given a finite set X (of elements called *points*) and integers $k, r, \lambda \geq 1$, we define a *2-design* (or *BIBD*, standing for balanced incomplete block design) B to be a family of k -element subsets of X , called *blocks*, such that any x in X is contained in r blocks, and any pair of distinct points x and y in X is contained in λ blocks. Here, the condition that any x in X is contained in r blocks is redundant, as shown below.

Here v (the number of elements of X , called points), b (the number of blocks), k , r , and λ are the *parameters* of the design. (To avoid degenerate examples, it is also assumed that $v > k$, so that no block contains all the elements of the set. This is the meaning of "incomplete" in the name of these designs.) In a table:

v	points, number of elements of X
-----	-----------------------------------

b	number of blocks
-----	------------------

r	number of blocks containing a given point
-----	---

k	number of points in a block
-----	-----------------------------

λ	number of blocks containing any 2 (or more generally t) distinct points
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The design is called a (v, k, λ) -design or a (v, b, r, k, λ) -design. The parameters are not all independent; v, k , and λ determine b and r , and not all combinations of v, k , and λ are possible. The two basic equations connecting these parameters are

$$bk = vr,$$

obtained by counting the number of pairs (B, p) where B is a block and p is a point in that block, and

$$\lambda(v-1) = r(k-1),$$

obtained from counting for a fixed x the triples (x, y, B) where x and y are distinct points and B is a block that contains them both. This equation for every x also proves that r is constant (independent of x) even without assuming it explicitly, thus proving that the condition that any x in X is contained in r blocks is redundant and r can be computed from the other parameters.

The resulting b and r must be integers, which imposes conditions on v, k , and λ . These conditions are not sufficient as, for example, a $(43, 7, 1)$ -design does not exist.

The *order* of a 2-design is defined to be $n = r - \lambda$. The **complement** of a 2-design is obtained by replacing each block with its complement in the point set X . It is also a 2-design and has parameters $v' = v, b' = b, r' = b - r, k' = v - k, \lambda' = \lambda + b - 2r$. A 2-design and its complement have the same order.

A fundamental theorem, [Fisher's inequality](#), named after the statistician [Ronald Fisher](#), is that $b \geq v$ in any 2-design.

A rather surprising and not very obvious (but very general) combinatorial result for these designs is that if points are denoted by any arbitrarily chosen set of equally or unequally spaced numerics, there is no choice of such a set which can make all block-sums (that is, sum of all points in a given block) constant. For other designs such as partially balanced incomplete block designs this may however be possible. Many such cases are discussed in. However, it can also be observed trivially for the magic squares or magic rectangles which can be viewed as the partially balanced incomplete block designs.

Examples

The unique $(6, 3, 2)$ -design ($v = 6, k = 3, \lambda = 2$) has 10 blocks ($b = 10$) and each element is repeated 5 times ($r = 5$). Using the symbols 0 – 5, the blocks are the following triples:

012 013 024 035 045 125 134 145 234 235.

and the corresponding [incidence matrix](#) (a $v \times b$ [binary matrix](#) with constant row sum r and constant column sum k) is:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

One of four nonisomorphic (8,4,3)-designs has 14 blocks with each element repeated 7 times. Using the symbols 0 – 7 the blocks are the following 4-tuples:

0123 0124 0156 0257 0345 0367 0467 1267 1346 1357 1457 2347
2356 2456.

The unique (7,3,1)-design is symmetric and has 7 blocks with each element repeated 3 times. Using the symbols 0 – 6, the blocks are the following triples:

013 026 045 124 156 235 346.

This design is associated with the [Fano plane](#), with the elements and blocks of the design [corresponding](#) to the points and lines of the plane. Its corresponding incidence matrix can also be symmetric, if the labels or blocks are sorted the right way:

```

1 1 1 0 0 0 0
1 0 0 1 1 0 0
1 0 0 0 0 1 1
( 0 1 0 1 0 1 0 )
0 1 0 0 1 0 1
0 0 1 1 0 0 1
0 0 1 0 1 1 0

```

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Symmetric 2-designs (SBIBDs)

The case of equality in Fisher's inequality, that is, a 2-design with an equal number of points and blocks, is called a **symmetric design**. Symmetric designs have the smallest number of blocks among all the 2-designs with the same number of points.

In a symmetric design $r = k$ holds as well as $b = v$, and, while it is generally not true in arbitrary 2-designs, in a symmetric design every two distinct blocks meet in λ points. A theorem of [Ryser](#) provides the converse. If X is a v -element set, and B is a v -element set of k -element subsets (the "blocks"), such that any two distinct blocks have exactly λ points in common, then (X, B) is a symmetric block design.

The parameters of a symmetric design satisfy

$$\lambda(v-1) = k(k-1).$$

This imposes strong restrictions on v , so the number of points is far from arbitrary. The [Bruck–Ryser–Chowla theorem](#) gives necessary, but not sufficient, conditions for the existence of a symmetric design in terms of these parameters.

The following are important examples of symmetric 2-designs:

Projective planes

Main article: [Projective plane](#)

[Finite projective planes](#) are symmetric 2-designs with $\lambda = 1$ and order $n > 1$. For these designs the symmetric design equation becomes:

$$v - 1 = k (k - 1) .$$

Since $k = r$ we can write the *order of a projective plane* as $n = k - 1$ and, from the displayed equation above, we obtain $v = (n + 1)n + 1 = n^2 + n + 1$ points in a projective plane of order n .

As a projective plane is a symmetric design, we have $b = v$, meaning that $b = n^2 + n + 1$ also. The number b is the number of *lines* of the projective plane. There can be no repeated lines since $\lambda = 1$, so a projective plane is a simple 2-design in which the number of lines and the number of points are always the same. For a projective plane, k is the number of points on each line and it is equal to $n + 1$. Similarly, $r = n + 1$ is the number of lines with which a given point is incident.

For $n = 2$ we get a projective plane of order 2, also called the [Fano plane](#), with $v = 4 + 2 + 1 = 7$ points and 7 lines. In the Fano plane, each line has $n + 1 = 3$ points and each point belongs to $n + 1 = 3$ lines.

Projective planes are known to exist for all orders which are prime numbers or powers of primes. They form the only known infinite family (with respect to having a constant λ value) of symmetric block designs.

Biplanes

A **biplane** or **biplane geometry** is a symmetric 2-design with $\lambda = 2$; that is, every set of two points is contained in two blocks ("lines"), while any two lines intersect in two points. They are similar to finite projective planes, except that rather than two points determining one line (and two lines determining one point), two points determine two lines (respectively, points). A biplane of order n is one whose blocks have $k = n + 2$ points; it has $v = 1 + (n + 2)(n + 1)/2$ points (since $r = k$).

The 18 known examples are listed below.

- (Trivial) The order 0 biplane has 2 points (and lines of size 2; a 2-(2,2,2) design); it is two points, with two blocks, each consisting of both points. Geometrically, it is the [digon](#).
- The order 1 biplane has 4 points (and lines of size 3; a 2-(4,3,2) design); it is the complete design with $v = 4$ and $k = 3$. Geometrically, the points are the vertices of a tetrahedron and the blocks are its faces.
- The order 2 biplane is the complement of the [Fano plane](#): it has 7 points (and lines of size 4; a 2-(7,4,2)), where the lines are given as the *complements* of the (3-point) lines in the Fano plane.
- The order 3 biplane has 11 points (and lines of size 5; a 2-(11,5,2)), and is also known as the **Paley biplane** after [Raymond Paley](#); it is associated to the [Paley digraph](#) of order 11, which is constructed using the field with 11 elements, and is associated to the size 12 Hadamard matrix; see [Paley construction I](#).

Algebraically this corresponds to the exceptional embedding of the [projective special linear group](#) $PSL(2,5)$ in $PSL(2,11)$ – see [projective linear group: action on \$p\$ points](#) for details.

- There are three biplanes of order 4 (and 16 points, lines of size 6; a 2-(16,6,2)). One is the [Kummer configuration](#). These three designs are also [Menon designs](#).
- There are four biplanes of order 7 (and 37 points, lines of size 9; a 2-(37,9,2)).
- There are five biplanes of order 9 (and 56 points, lines of size 11; a 2-(56,11,2)).
- Two biplanes are known of order 11 (and 79 points, lines of size 13; a 2-(79,13,2)).

Biplanes of orders 5, 6, 8 and 10 do not exist, as shown by the [Bruck-Ryser-Chowla theorem](#).

Hadamard 2-designs

An [Hadamard matrix](#) of size m is an $m \times m$ matrix $\mathbf{H}$ whose entries are ± 1 such that $\mathbf{H}\mathbf{H}^T = m\mathbf{I}_m$, where $\mathbf{H}^T$ is the transpose of $\mathbf{H}$ and $\mathbf{I}_m$ is the $m \times m$ identity matrix. An Hadamard matrix can be put into *standardized form* (that is, converted to an equivalent Hadamard matrix) where the first row and first column entries are all +1. If the size $m > 2$ then m must be a multiple of 4.

Given an Hadamard matrix of size $4a$ in standardized form, remove the first row and first column and convert every -1 to a 0. The resulting 0–1 matrix $\mathbf{M}$ is the [incidence matrix](#) of a symmetric 2-($4a - 1, 2a - 1, a - 1$) design called an **Hadamard 2-design**. It contains $4a - 1$ blocks/points; each contains/is contained in $2a - 1$ points/blocks. Each pair of points is contained in exactly $a - 1$ blocks.

This construction is reversible, and the incidence matrix of a symmetric 2-design with these parameters can be used to form an Hadamard matrix of size $4a$.

Resolvable 2-designs

A **resolvable 2-design** is a BIBD whose blocks can be partitioned into sets (called *parallel classes*), each of which forms a partition of the point set of the BIBD. The set of parallel classes is called a *resolution* of the design.

If a 2-(v, k, λ) resolvable design has c parallel classes, then $b \geq v + c - 1$.

Consequently, a symmetric design can not have a non-trivial (more than one parallel class) resolution.

Archetypical resolvable 2-designs are the finite [affine planes](#). A solution of the famous [15 schoolgirl problem](#) is a resolution of a 2-(15,3,1) design.

General balanced designs (t -designs)

Given any positive integer t , a t -design B is a class of k -element subsets of X , called *blocks*, such that every point x in X appears in exactly r blocks, and every t -element subset T appears in exactly λ blocks. The numbers v (the number of elements of X), b (the number of blocks), k , r , λ , and t are the *parameters* of the design. The design may be called a t -(v, k, λ)-design. Again, these four numbers determine b and r and the four numbers themselves cannot be chosen arbitrarily. The equations are

$$\lambda_i = \lambda \binom{v-i}{t-i} / \binom{k-i}{t-i} \text{ for } i = 0, 1, \dots, t,$$

$$\lambda_i = \lambda \binom{v-i}{t-i} / \binom{k-i}{t-i} \text{ for } i = 0, 1, \dots, t,$$

where λ_i is the number of blocks that contain any i -element set of points and $\lambda_t = \lambda$.

Note that $b = \lambda_0 = \lambda \binom{v}{t} / \binom{k}{t}$

$$b = \lambda_0 = \lambda \binom{v}{t} / \binom{k}{t}$$

and $r = \lambda_1 = \lambda \binom{v-1}{t-1} / \binom{k-1}{t-1}$

$$r = \lambda_1 = \lambda \binom{v-1}{t-1} / \binom{k-1}{t-1}$$

Theorem: Any t -(v, k, λ)-design is also an s -(v, k, λ_s)-design for any s with $1 \leq s \leq t$. (Note that the "lambda value" changes as above and depends on s .)

A consequence of this theorem is that every t -design with $t \geq 2$ is also a 2-design.

A t -($v, k, 1$)-design is called a [Steiner system](#).

The term *block design* by itself usually means a 2-design.

Derived and extendable t -designs

Let $\mathbf{D} = (X, B)$ be a t -(v, k, λ) design and p a point of X . The *derived design* D_p has point set $X - \{p\}$ and as block set all the blocks of $\mathbf{D}$ which contain p with p removed. It is a $(t-1)$ -($v-1, k-1, \lambda$) design. Note that derived designs with respect to different points may not be isomorphic. A design $\mathbf{E}$ is called an *extension* of $\mathbf{D}$ if $\mathbf{E}$ has a point p such that $\mathbf{E}_p$ is isomorphic to $\mathbf{D}$; we call $\mathbf{D}$ *extendable* if it has an extension.

Theorem: If a t -(v, k, λ) design has an extension, then $k+1$ divides $b(v+1)$.

The only extendable [projective planes](#) (symmetric 2-($n^2 + n + 1, n + 1, 1$) designs) are those of orders 2 and 4.

Every Hadamard 2-design is extendable (to an **Hadamard 3-design**).

Theorem: If $\mathbf{D}$, a symmetric 2-(v, k, λ) design, is extendable, then one of the following holds:

1. $\mathbf{D}$ is an Hadamard 2-design,
2. $v = (\lambda + 2)(\lambda^2 + 4\lambda + 2)$, $k = \lambda^2 + 3\lambda + 1$,
3. $v = 495$, $k = 39$, $\lambda = 3$.

Note that the projective plane of order two is an Hadamard 2-design; the projective plane of order four has parameters which fall in case 2; the only other known symmetric 2-designs with parameters in case 2 are the order 9 biplanes, but none of them are extendable; and there is no known symmetric 2-design with the parameters of case 3.

Inversive planes

A design with the parameters of the extension of an [affine plane](#), i.e., a $3-(n^2 + 1, n + 1, 1)$ design, is called a finite **inversive plane**, or [Möbius plane](#), of order n .

It is possible to give a geometric description of some inversive planes, indeed, of all known inversive planes. An [ovoid](#) in $\text{PG}(3, q)$ is a set of $q^2 + 1$ points, no three collinear. It can be shown that every plane (which is a hyperplane since the geometric dimension is 3) of $\text{PG}(3, q)$ meets an ovoid O in either 1 or $q + 1$ points. The plane sections of size $q + 1$ of O are the blocks of an inversive plane of order q . Any inversive plane arising this way is called *egglike*. All known inversive planes are egglike.

An example of an ovoid is the [elliptic quadric](#), the set of zeros of the quadratic form

$$x_1x_2 + f(x_3, x_4),$$

where f is an irreducible [quadratic form](#) in two variables over $\text{GF}(q)$. [$f(x, y) = x^2 + xy + y^2$ for example].

If q is an odd power of 2, another type of ovoid is known – the [Suzuki–Tits ovoid](#).

Theorem. Let q be a positive integer, at least 2. (a) If q is odd, then any ovoid is projectively equivalent to the elliptic quadric in a projective geometry $\text{PG}(3, q)$; so q is a prime power and there is a unique egglike inversive plane of order q . (But it is unknown if non-egglike ones exist.) (b) if q is even, then q is a power of 2 and any inversive plane of order q is egglike (but there may be some unknown ovoids).

Partially balanced designs (PBIBDs)

An n -class [association scheme](#) consists of a [set](#) X of size v together with a [partition](#) S of $X \times X$ into $n + 1$ [binary relations](#), $R_0, R_1, \dots, R_n$. A pair of elements in relation R_i are said to be i th-*associates*. Each element of X has n_i i th associates. Furthermore:

- $R_0 = \{ (x, x) : x \in X \}$ and is called the [Identity relation](#).
- Defining $R^* := \{ (x, y) \mid (y, x) \in R \}$, if R in S , then R^* in S
- If $(x, y) \in R_k$, the number of $z \in X$ such that $(x, z) \in R_i$ and $(z, y) \in R_j$ is a constant p_{ij}^k depending on i, j, k but not on the particular choice of x and y .

An association scheme is *commutative* if $p_{ij}^k = p_{ji}^k$ for all i, j and k . Most authors assume this property.

A **partially balanced incomplete block design** with n associate classes (PBIBD(n)) is a block design based on a v -set X with b blocks each of size k and with each element appearing in r blocks, such that there is an association scheme with n classes defined on X where, if elements x and y are i th associates, $1 \leq i \leq n$, then they are together in precisely λ_i blocks.

A PBIBD(n) determines an association scheme but the converse is false.

Example

Let $A(3)$ be the following association scheme with three associate classes on the set $X = \{1, 2, 3, 4, 5, 6\}$. The (i, j) entry is s if elements i and j are in relation R_s .

	1	2	3	4	5	6
1	0	1	1	2	3	3
2	1	0	1	3	2	3
3	1	1	0	3	3	2
4	2	3	3	0	1	1
5	3	2	3	1	0	1
6	3	3	2	1	1	0

The blocks of a PBIBD(3) based on $A(3)$ are:

124	134	235	456
125	136	236	456

The parameters of this PBIBD(3) are: $v = 6$, $b = 8$, $k = 3$, $r = 4$ and $\lambda_1 = \lambda_2 = 2$ and $\lambda_3 = 1$. Also, for the association scheme we have $n_0 = n_2 = 1$ and $n_1 = n_3 = 2$. The incidence matrix M is

$$\begin{pmatrix}
 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\
 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\
 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\
 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\
 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\
 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1
 \end{pmatrix}$$

$$\begin{pmatrix}
 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\
 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\
 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\
 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\
 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\
 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1
 \end{pmatrix}$$

and the concurrence matrix MM^T is

$$\begin{pmatrix}
 4 & 2 & 2 & 2 & 1 & 1 \\
 2 & 4 & 2 & 1 & 2 & 1 \\
 2 & 2 & 4 & 1 & 1 & 2 \\
 2 & 1 & 1 & 4 & 2 & 2 \\
 1 & 2 & 1 & 2 & 4 & 2 \\
 1 & 1 & 2 & 2 & 2 & 4
 \end{pmatrix}$$

$$\begin{pmatrix}
 4 & 2 & 2 & 2 & 1 & 1 \\
 2 & 4 & 2 & 1 & 2 & 1 \\
 2 & 2 & 4 & 1 & 1 & 2 \\
 2 & 1 & 1 & 4 & 2 & 2 \\
 1 & 2 & 1 & 2 & 4 & 2 \\
 1 & 1 & 2 & 2 & 2 & 4
 \end{pmatrix}$$

from which we can recover the λ and r values.

Properties

The parameters of a PBIBD(m) satisfy:

1. $vr = bk$

$$2. \sum_{i=1}^m n_i = v - 1$$

$$\sum_{i=1}^m n_i = v - 1$$

$$3. \sum_{i=1}^m n_i \lambda_i = r (k - 1)$$

$$\sum_{i=1}^m n_i \lambda_i = r(k - 1)$$

$$4. \sum_{u=0}^m p_{ju}^h = n_j$$

$$\sum_{u=0}^m p_{ju}^h = n_j$$

$$5. n_i p_{jh}^i = n_j p_{ih}^j$$

A PBIBD(1) is a BIBD and a PBIBD(2) in which $\lambda_1 = \lambda_2$ is a BIBD.

Two associate class PBIBDs

PBIBD(2)s have been studied the most since they are the simplest and most useful of the PBIBDs. They fall into six types based on a classification of the *then known* PBIBD(2)s by :

1. group divisible;
2. triangular;
3. Latin square type;
4. cyclic;
5. partial geometry type;
6. miscellaneous.

Applications

The mathematical subject of block designs originated in the statistical framework of [design of experiments](#). These designs were especially useful in applications of the technique of [analysis of variance \(ANOVA\)](#). This remains a significant area for the use of block designs.

While the origins of the subject are grounded in biological applications (as is some of the existing terminology), the designs are used in many applications where systematic comparisons are being made, such as in [software testing](#).

The [incidence matrix](#) of block designs provide a natural source of interesting [block codes](#) that are used as [error correcting codes](#). The rows of their incidence matrices are also used as the symbols in a form of [pulse-position modulation](#).

Statistical application

Suppose that skin cancer researchers want to test three different sunscreens. They coat two different sunscreens on the upper sides of the hands of a test person. After a UV radiation they record the skin irritation in terms of sunburn. The number of treatments is 3 (sunscreens) and the block size is 2 (hands per person).

A corresponding BIBD can be generated by the [R-function](#) *design.bib* of the [R-package agricolae](#) and is specified in the following table:

Plots	Block	Treatment
101	1	3
102	1	2
201	2	1
202	2	3
301	3	2
302	3	1

The investigator chooses the parameters $v = 3$, $k = 2$ and $\lambda = 1$ for the block design which are then inserted into the R-function. Subsequently, the remaining parameters b and r are determined automatically.

Using the basic relations we calculate that we need $b = 3$ blocks, that is, 3 test people in order to obtain a balanced incomplete block design. Labeling the blocks A , B and C , to avoid confusion, we have the block design,

$A = \{2, 3\}$, $B = \{1, 3\}$ and $C = \{1, 2\}$.

A corresponding [incidence matrix](#) is specified in the following table:

Treatment	Block A	Block B	Block C
1	0	1	1
2	1	0	1
3	1	1	0

Each treatment occurs in 2 blocks, so $r = 2$.

Just one block (C) contains the treatments 1 and 2 simultaneously and the same applies to the pairs of treatments (1,3) and (2,3). Therefore, $\lambda = 1$.

It is impossible to use a complete design (all treatments in each block) in this example because there are 3 sunscreens to test, but only 2 hands on each person.

See also

- [Incidence geometry](#)
- [Steiner system](#)
- [Fractional factorial design](#)

Notes

1. , pp.17–19
2. , p.1
3. P. Dobcsányi, D.A. Preece. L.H. Soicher (2007-10-01). "[On balanced incomplete-block designs with repeated blocks](#)". [European Journal of Combinatorics](#). **28** (7): 1955–1970. [doi:10.1016/j.ejc.2006.08.007](#). [ISSN 0195-6698](#).
4. Proved by Tarry in 1900 who showed that there was no pair of orthogonal [Latin squares](#) of order six. The 2-design with the indicated parameters is equivalent to the existence of five mutually orthogonal Latin squares of order six.
5. ^ , p. 27
6. They have also been referred to as *projective designs* or *square designs*. These alternatives have been used in an attempt to replace the term "symmetric", since there is nothing symmetric (in the usual meaning of the term) about these designs. The use of *projective* is due to P.Dembowski (*Finite Geometries*, Springer, 1968), in analogy with the most common example, projective planes, while *square* is due to P. Cameron (*Designs, Graphs, Codes and their Links*, Cambridge, 1991) and captures the implication of $v = b$ on the incidence matrix. Neither term has caught on as a replacement and these designs are still universally referred to as *symmetric*.
7. , pg.23, Theorem 2.2
8. , pp. 102–104
9. ^ , pg.109
10. , pp.320–335
11. , pg.55
12. Martin, Pablo; Singerman, David (April 17, 2008), [From Biplanes to the Klein quartic and the Buckyball](#) (PDF), p. 4
13. , pp. 279–281
14. , pg. 74, Theorem 4.5
15. , pg. 156, Theorem 5.4
16. , pg. 158, Corollary 5.5
17. , pg. 40 Example 5.8
18. , pg.203, Corollary 9.6
19. , pg.29

20. , pg. 11, Proposition 1.34
21. , pg. 132, Theorem 4.5
22. , pg. 11, Theorem 1.35
23. , pg. 114, Remarks 6.35
24. , pg. 237
25. , pg. 238
26. , pg. 240, Lemma 4
27. , pg. 562, Remark 42.3 (4)
28. , pg. 242
29. Not a mathematical classification since one of the types is a catch-all "and everything else".
30. , pg. 127
31. Noshad, Mohammad; [Brandt-Pearce, Maïté](#) (Jul 2012). "Expurgated PPM Using Symmetric Balanced Incomplete Block Designs". *IEEE Communications Letters*. **16** (7): 968–971. [arXiv:1203.5378](#). [Bibcode:2012arXiv1203.5378N](#). [doi:10.1109/LCOMM.2012.042512.120457](#). [S2CID 7586742](#).

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External links

- [DesignTheory.Org](#): Databases of combinatorial, statistical, and experimental block designs. Software and other resources hosted by the School of Mathematical Sciences at Queen Mary College, University of London.
- [Design Theory Resources](#): [Peter Cameron](#)'s page of web based design theory resources.
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Incidence structures

- | | |
|-----------------------|---|
| Representation | <ul style="list-style-type: none"> • Incidence matrix • Incidence graph |
|-----------------------|---|
-

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